

# Slope-Intercept Form ( $y = mx + b$ )

*Problems matched to Math with Mr. J's tutorial — with graphs*

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Class: \_\_\_\_\_

## 1. Identify slope (m) and y-intercept (b) from the equation

In  $y = mx + b$ , the number in front of  $x$  is the slope; the constant is the y-intercept.

①  $y = 2x + 3$                        $\rightarrow m =$  \_\_\_\_\_  $b =$  \_\_\_\_\_

②  $y = -4x + 7$                        $\rightarrow m =$  \_\_\_\_\_  $b =$  \_\_\_\_\_

③  $y = (1/2)x - 1$                        $\rightarrow m =$  \_\_\_\_\_  $b =$  \_\_\_\_\_

④  $y = -(2/3)x + 5$                        $\rightarrow m =$  \_\_\_\_\_  $b =$  \_\_\_\_\_

⑤  $y = x + 6$                        $\rightarrow m =$  \_\_\_\_\_  $b =$  \_\_\_\_\_

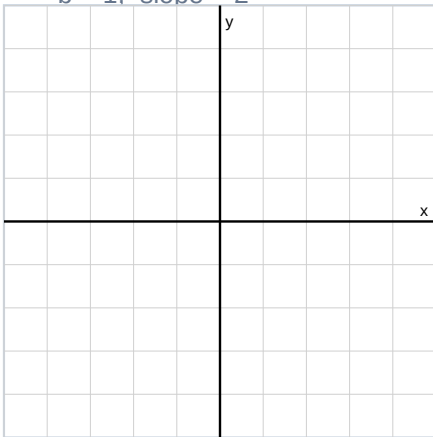
⑥  $y = -3x$                        $\rightarrow m =$  \_\_\_\_\_  $b =$  \_\_\_\_\_

### 2. Graph each line — start at $b$ , then use the slope

Plot the  $y$ -intercept  $(0, b)$  first. Then use slope = rise/run to plot a second point.

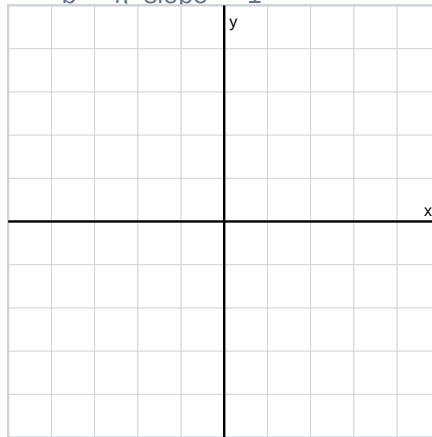
⑦  $y = 2x + 1$

$b = 1$ , slope = 2



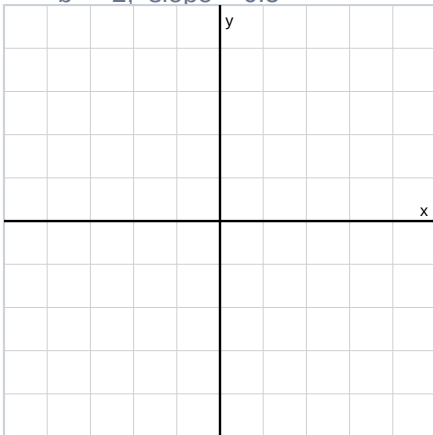
⑧  $y = -x + 4$

$b = 4$ , slope = -1



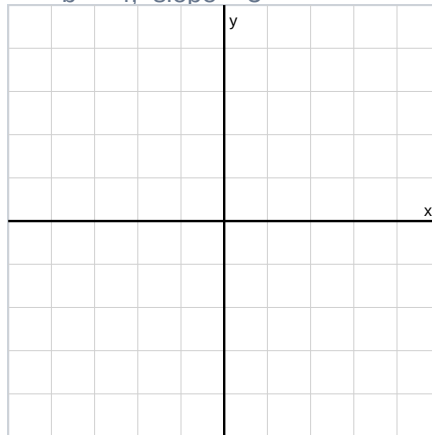
⑨  $y = (1/2)x - 2$

$b = -2$ , slope = 0.5



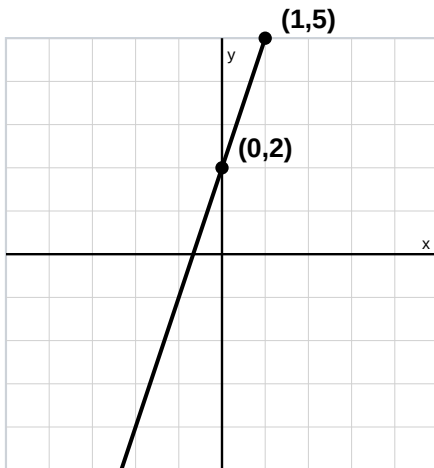
⑩  $y = 3x - 4$

$b = -4$ , slope = 3



### 3. Write the equation from the graph

Find where the line crosses the y-axis (that's  $b$ ). Count rise/run for  $m$ . Write  $y = mx + b$ .

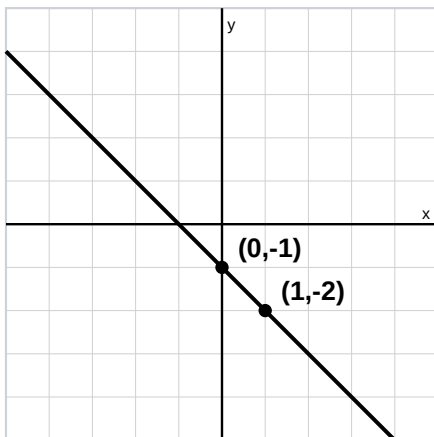


⑪ Read the graph and write:

$$m = \underline{\hspace{2cm}}$$

$$b = \underline{\hspace{2cm}}$$

$$y = \underline{\hspace{4cm}}$$

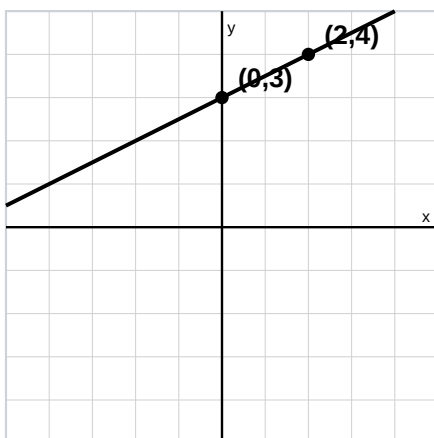


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⑬ Read the graph and write:

$$m = \underline{\hspace{2cm}}$$

$$b = \underline{\hspace{2cm}}$$

$$y = \underline{\hspace{4cm}}$$

### 4. Rewrite in slope-intercept form (solve for y)

Get y by itself. Move x-term to the other side, divide if needed.

⑭  $3x + y = 5$        $\rightarrow y =$  \_\_\_\_\_       $m =$  \_\_\_\_\_       $b =$  \_\_\_\_\_

⑮  $2x - y = 4$        $\rightarrow y =$  \_\_\_\_\_       $m =$  \_\_\_\_\_       $b =$  \_\_\_\_\_

⑯  $x + 2y = 6$        $\rightarrow y =$  \_\_\_\_\_       $m =$  \_\_\_\_\_       $b =$  \_\_\_\_\_

⑰  $-4x + 2y = 8$        $\rightarrow y =$  \_\_\_\_\_       $m =$  \_\_\_\_\_       $b =$  \_\_\_\_\_